

Problems for week 8

(Questions on this problem sheet are NOT assessed)

Problem 1. Suppose $\vec{u}, \vec{v} : \mathbb{R} \rightarrow \mathbb{R}^3$ are differentiable vector functions. Show that:

$$\frac{d}{dt}(\vec{u}(t) \cdot \vec{v}(t)) = \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t)$$

Problem 2. Suppose $\vec{u}, \vec{v} : \mathbb{R} \rightarrow \mathbb{R}^3$ are differentiable vector functions. Show that:

$$\frac{d}{dt}(\vec{u}(t) \times \vec{v}(t)) = \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t)$$

Problem 3. Consider the vector field

$$\vec{F} = (x + y^2)\hat{i} + xz\hat{j} + (y + z)\hat{k}.$$

Evaluate the line integral

$$\int_C \vec{F} \cdot d\vec{r},$$

where the path C is given in terms of its position vector by

$$\vec{r}(t) = t^2\hat{i} + t^3\hat{j} - 2t\hat{k}, \quad 0 \leq t \leq 2.$$

Problem 4. The force exerted by an electric charge at the origin on a charged particle at a point (x, y, z) with position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is $\vec{F}(\vec{r}) = K\vec{r}/|\vec{r}|^3$ where K is a constant. Compute the work done as the particle moves along a straight line from $(2, 0, 0)$ to $(2, 1, 5)$.

Problem 5. Consider the vector field $\vec{F} = yz\hat{i} + xz\hat{j} + (xy + 2z)\hat{k}$, and let C be the line segment from $(1, 0, -2)$ to $(4, 6, 3)$.

- Find a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that $\vec{F} = \nabla f$. Show your work.
- Use part (a) to evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$.

Problem 6. Consider the vector field $\vec{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, defined as follows:

$$\vec{F}(x, y) = \frac{-y\hat{i} + x\hat{j}}{x^2 + y^2}.$$

- (a) Show that $\vec{F}$ is the gradient of the polar angle function $\theta(x, y) = \tan^{-1}(\frac{y}{x})$ defined over the right half-plane $x > 0$. (Note: this formula for θ does not make sense for $x = 0$!)
- (b) Suppose that C is a smooth planar curve in the right half-plane $\{(x, y) \in \mathbb{R}^2 : x > 0\}$ joining two points $A = (x_1, y_1)$ and $B = (x_2, y_2)$. Express the line integral

$$\int_C \vec{F} \cdot d\vec{r}$$

in terms of the polar coordinates (r_1, θ_1) and (r_2, θ_2) of A and B .

- (c) Compute directly from the definition the line integrals

$$\int_{C_1} \vec{F} \cdot d\vec{r} \text{ and } \int_{C_2} \vec{F} \cdot d\vec{r},$$

where C_1 is the upper half of the unit circle running from $(1, 0)$ to $(-1, 0)$, and C_2 is the lower half of the unit circle, also going from $(1, 0)$ to $(-1, 0)$.

- (d) Using the results of parts (a)–(c), is the vector field $\vec{F}$ conservative over its entire domain of definition? Is it conservative over the right half-plane $x > 0$? Justify your answers.